Ideal Fluid Flow in a Polygonal Domain via the Schwarz–Christoffel Transformation

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Abstract

This paper applies the Schwarz–Christoffel (SC) conformal mapping to simulate steady, irrotational, incompressible ideal fluid flow inside a 12-vertex simplification of the Boulder County, Colorado boundary polygon. The SC transformation maps the upper half-plane $\mathbb{H}$ conformally onto the polygonal domain Ω, converting analytically tractable complex potentials in $\mathbb{H}$ into streamline and equipotential patterns in the physical domain. Two flow models are derived and implemented: a uniform ideal flow $W = U\zeta$ providing the baseline conformal grid, and a doubly-connected flow in which a downtown obstacle is modelled via the Milne-Thomson circle theorem. The SC parameter problem is solved by softmax pre-vertex parameterisation combined with Levenberg–Marquardt least squares on side-length ratios computed by Gauss–Legendre quadrature. All results are produced through the forward SC map, yielding streamlines that satisfy the no-penetration boundary condition exactly by construction. A closed-form verification on the rectangle-to-half-plane map confirms the implementation.

1 Introduction

Computing fluid flow in an arbitrary domain is a hard problem: Laplace's equation $\nabla^2\phi = 0$ admits no elementary closed-form solution in an irregular geometry. The classical approach uses conformal mapping. A conformal map $z = f(\zeta)$ from a canonical domain (the unit disk or the upper half-plane) to the physical region Ω transforms the Laplace equation from one domain to the other while preserving harmonicity, so that any analytic function $W(\zeta)$ satisfying the required boundary conditions in the canonical domain produces a valid solution in Ω via $\tilde{W}(z) = W(f^{-1}(z))$.

The Schwarz–Christoffel (SC) transformation, developed independently by H. A. Schwarz and E. B. Christoffel in the 1860s, gives an explicit integral formula for the conformal map from the upper half-plane $\mathbb{H}$ to any simply connected polygon. For analytic polygons the parameter problem (choosing pre-images of the vertices on the real axis) can sometimes be solved in closed form; for general n-gons it must be treated numerically. The modern numerical theory was systematised by Driscoll and Trefethen [Driscoll and Trefethen, 2002]; the classical theory is in Nehari [Nehari, 1952], Ahlfors [Ahlfors, 1979], and Walker [Walker, 1964].

We apply the SC transformation to a 12-vertex polygon derived from the Boulder County, Colorado boundary, chosen for its mix of convex and reflex vertices as a non-trivial test of the numerical pipeline. Two flow regimes are considered: (i) uniform ideal flow, providing the baseline conformal grid, and (ii) doubly-connected flow with an interior obstacle modelled via the Milne-Thomson circle theorem. The polygon's geometric complexity exercises the full pipeline; any physical interpretation as "airflow over a city" is illustrative. The mathematical framework is drawn from Ablowitz and Fokas [2003], Driscoll and Trefethen [2002], Nehari [1952]; the fluid-dynamics background from Bender and Orszag [1999] and Milne-Thomson [1968].

A closed-form verification against the rectangle-to-half-plane map (an elliptic-integral identity) is included to certify the numerical pipeline independently of the Boulder application.

Organisation. Section 2 develops the SC mapping theory and complex-potential framework, with step-by-step derivations of every key result. Section 3 describes the numerical algorithms and verifies the pipeline against the closed-form rectangle map. Section 4 applies the method to the Boulder polygon and presents results for both flow models. Section 5 summarises the conclusions. Technical details on the Levenberg–Marquardt solver are in Appendix A; code availability in Appendix B.

2 Mathematical Framework

2.1 Ideal Fluid Flow and the Complex Potential

A steady two-dimensional flow is *ideal* when the velocity field $\mathbf{u} = (u, v)$ is simultaneously irrotational ($\partial_x v - \partial_y u = 0$) and incompressible ($\partial_x u + \partial_y v = 0$). Irrotationality in a simply connected region forces $\mathbf{u}$ to be a gradient field $\mathbf{u} = \nabla\phi$ (velocity potential), and incompressibility then gives $\nabla^2\phi = 0$. Incompressibility independently forces $\mathbf{u}$ to be the skew-gradient of a stream function ψ via $u = \partial_y\psi$, $v = -\partial_x\psi$, and irrotationality then gives $\nabla^2\psi = 0$. Comparing the two representations produces the Cauchy–Riemann equations

$$\partial_x\phi = \partial_y\psi, \qquad \partial_y\phi = -\partial_x\psi, \tag{1}$$

so the *complex potential*

$$W(z) = \phi(x, y) + i\,\psi(x, y), \qquad z = x + iy, \tag{2}$$

is analytic on the flow domain.

Velocity from $W'(z)$. For any analytic function the complex derivative can be evaluated along the real direction: $W'(z) = \partial_x W = \partial_x \phi + i\,\partial_x \psi$. Using $\partial_x \phi = u$ and the Cauchy–Riemann equation $\partial_x \psi = -\partial_y \phi = -v$, we obtain

$$W'(z) = u - iv = \overline{u + iv}, \qquad \mathbf{u} = \overline{W'(z)}. \tag{3}$$

Physical interpretation. Level curves of ϕ are *equipotentials*; level curves of ψ are *streamlines*. The Cauchy–Riemann equations imply $\nabla\phi \cdot \nabla\psi = 0$, so streamlines and equipotentials meet orthogonally wherever $W'(z) \neq 0$. The no-penetration condition on a solid boundary $\partial\Omega$ reads $\mathbf{u} \cdot \hat{\mathbf{n}} = 0$, equivalent to $\psi = \text{const}$ along $\partial\Omega$.

2.2 Conformal Invariance of Harmonicity

Lemma 1. *Let $f : \mathbb{H} \to \Omega$ be a bijective analytic map with $f'(\zeta) \neq 0$. If $W(\zeta)$ is analytic on $\mathbb{H}$, then $\tilde{W}(z) = W(f^{-1}(z))$ is analytic on Ω; in particular, both $\tilde{\phi} = \mathrm{Re}\tilde{W}$ and $\tilde{\psi} = \mathrm{Im}\tilde{W}$ satisfy Laplace's equation on Ω.*

This is the structural fact that makes conformal mapping work for ideal flow. Any solution of Laplace's equation in $\mathbb{H}$ pulls back to a solution in Ω, and the no-penetration condition $\psi = 0$ on $\partial\mathbb{H}$ (easy to arrange by the method of images) becomes $\psi = 0$ on $\partial\Omega$ automatically.

2.3 The Riemann Mapping Theorem

Theorem 2 (Riemann, 1851)**.** *Let $\Omega \subsetneq \mathbb{C}$ be a simply connected domain. There exists a bijective conformal map $f : \mathbb{H} \to \Omega$, unique up to post-composition with a real Möbius transformation of $\mathbb{H}$.*

A proof is in Ahlfors [1979, Ch. 6]. The Boulder polygon is simply connected, so f exists. The three real parameters of the Möbius freedom correspond to fixing three real pre-images on $\partial\mathbb{H} = \mathbb{R} \cup \{\infty\}$.

2.4 The Schwarz–Christoffel Formula

For a polygon Ω with vertices $w_1, \ldots, w_n$ (counter-clockwise) and interior angles $\alpha_k \pi$, $\alpha_k \in (0, 2)$, the SC map takes the form [Nehari, 1952, Ch. V, §6][Ahlfors, 1979, Ch. 6, §2.2][Driscoll and Trefethen, 2002, Ch. 2]

$$f(\zeta) = A + C \int_{\zeta_0}^{\zeta} \prod_{k=1}^{n} (t - \zeta_k)^{\alpha_k - 1}\,\mathrm{d}t, \tag{4}$$

where $\zeta_k \in \mathbb{R}$ are the *pre-vertices*, $A \in \mathbb{C}$ is a translation, and $C \in \mathbb{C}$ encodes scale and rotation. The angle-sum identity $\sum_k \alpha_k = n - 2$ follows from the total CCW exterior turning $= 2\pi$, and convergence at infinity (when $\zeta_n = \infty$) holds whenever $\alpha_n > 0$.

The structural argument that produces (4) computes $\arg f'(t)$ for real $t \in (\zeta_k, \zeta_{k+1})$. Factors $(t - \zeta_j)^{\alpha_j - 1}$ with $j \leq k$ contribute zero phase; those with $j > k$ contribute $(\alpha_j - 1)\pi$ on the principal branch. Therefore

$$\arg f'(t) = \arg C + \pi \sum_{j=k+1}^{n} (\alpha_j - 1) \tag{5}$$

is constant on each open interval (ζ_k, ζ_{k+1}), so its image is a straight segment. As t crosses ζ_k, the tangent turns CCW by $(1-\alpha_k)\pi$, reproducing the prescribed exterior angle and hence the interior angle $\alpha_k \pi$.

Interior-angle formula in practice. Given vertices $w_0, \ldots, w_{n-1}$ (CCW), the signed exterior turning at w_k is $\arg\bigl((w_{k+1}-w_k)/(w_k - w_{k-1})\bigr)$, so

$$\alpha_k = 1 - \frac{1}{\pi} \arg\left(\frac{w_{k+1}-w_k}{w_k - w_{k-1}}\right), \tag{6}$$

with indices modulo n.

2.5 Möbius Normalisation and the Parameter Problem

The real Möbius group $\mathrm{PSL}(2,\mathbb{R})$, acting on $\mathbb{H}$ via $\zeta \mapsto (a\zeta + b)/(c\zeta + d)$, has three real parameters. Three of the n pre-vertices can therefore be fixed at prescribed values without loss of generality. We use

$$\zeta_0 = -1, \qquad \zeta_1 = 0, \qquad \zeta_{n-1} = 1, \tag{7}$$

leaving $n-3$ free parameters $\zeta_2 < \cdots < \zeta_{n-2}$ in $(0,1)$. For Boulder County, $n = 12$ yields $n - 3 = 9$ free pre-vertices. These are determined by matching side-length ratios; see Section 3.2.

2.6 Method of Images

Any analytic $W(\zeta)$ with $\mathrm{Im} W = 0$ on $\mathbb{R}$ automatically satisfies $\psi = 0$ on $\partial\Omega$ after pull-back via f. To add a singularity at $s = a + ib \in \mathbb{H}$ (with $b > 0$) while preserving $\mathrm{Im} W = 0$ on the real axis, we pair it with a mirror-image singularity at $\bar{s} = a - ib$. For a source of strength $Q > 0$ the image-pair potential is

$$W_{\text{source}}(\zeta) = \frac{Q}{2\pi}\bigl[\log(\zeta - s) + \log(\zeta - \bar{s})\bigr]. \tag{8}$$

Algebraic verification. Take $\zeta = x \in \mathbb{R}$:

$$\log(x-s) + \log(x - \bar{s}) = \log\bigl[(x-a)^2 + b^2\bigr] + i\,\bigl[\arg(x - a - ib) + \arg(x - a + ib)\bigr].$$

The numbers $x - a - ib$ and $x - a + ib$ are complex conjugates, so their arguments are opposite in sign, making the bracketed imaginary part vanish. Therefore

$$\mathrm{Im}\bigl[\log(x-s) + \log(x-\bar{s})\bigr] = 0 \quad \text{for all } x \in \mathbb{R}. \tag{9}$$

2.7 Milne-Thomson Circle Theorem

Theorem 3 (Milne-Thomson, 1940)**.** *Let $W_0(\zeta)$ be analytic on $\mathbb{C} \setminus D$ where $D = \{\zeta : |\zeta - \zeta_0| \le a\}$, with no singularities on ∂D. Then*

$$W^{\text{ext}}(\zeta) = W_0(\zeta) + \overline{W_0\left(\zeta_0 + \frac{a^2}{\overline{\zeta - \zeta_0}}\right)} \tag{10}$$

is analytic on the exterior of D with the same singularities as W_0, and satisfies $\mathrm{Im} W^{\text{ext}} = 0$ *on ∂D.*

Proof. Writing $\zeta - \zeta_0 = ae^{i\theta}$ on ∂D gives $a^2/\overline{\zeta - \zeta_0} = \zeta - \zeta_0$, so the inversion argument in (10) collapses to ζ itself and $W^{\text{ext}}|_{\partial D} = W_0(\zeta) + \overline{W_0(\zeta)} = 2\mathrm{Re}W_0(\zeta) \in \mathbb{R}$. Analyticity of the second term in ζ on $\mathbb{C} \setminus \overline{D}$ follows from the Schwarz reflection principle for circles [Ahlfors, 1979, Ch. 4, §6.5]. □

Upper-half-plane specialisation. Equation (10) enforces ∂D as a streamline on $\mathbb{C} \setminus D$. To additionally satisfy $\psi = 0$ on $\partial\mathbb{H}$, the method of images requires reflecting the obstacle dipole to $\overline{\zeta_0}$. For $W_0 = U\zeta$ the combined expression

$$W_{\mathrm{urban}}(\zeta) = U\zeta + \frac{Ua^2}{\zeta - \zeta_0} + \frac{Ua^2}{\zeta - \overline{\zeta_0}} \tag{11}$$

is holomorphic in ζ. The image dipole at $\overline{\zeta_0}$ enforces $\psi = 0$ on $\partial\mathbb{H}$ exactly (conjugate-pair cancellation as in Section 2.6); the dipole at ζ_0 approximates no-penetration on ∂D to leading order in $(a/\operatorname{Im}\zeta_0)^2$ [Milne-Thomson, 1968, §6.21], an order-of-magnitude estimate rather than a rigorous bound. Figure 1 illustrates the construction.

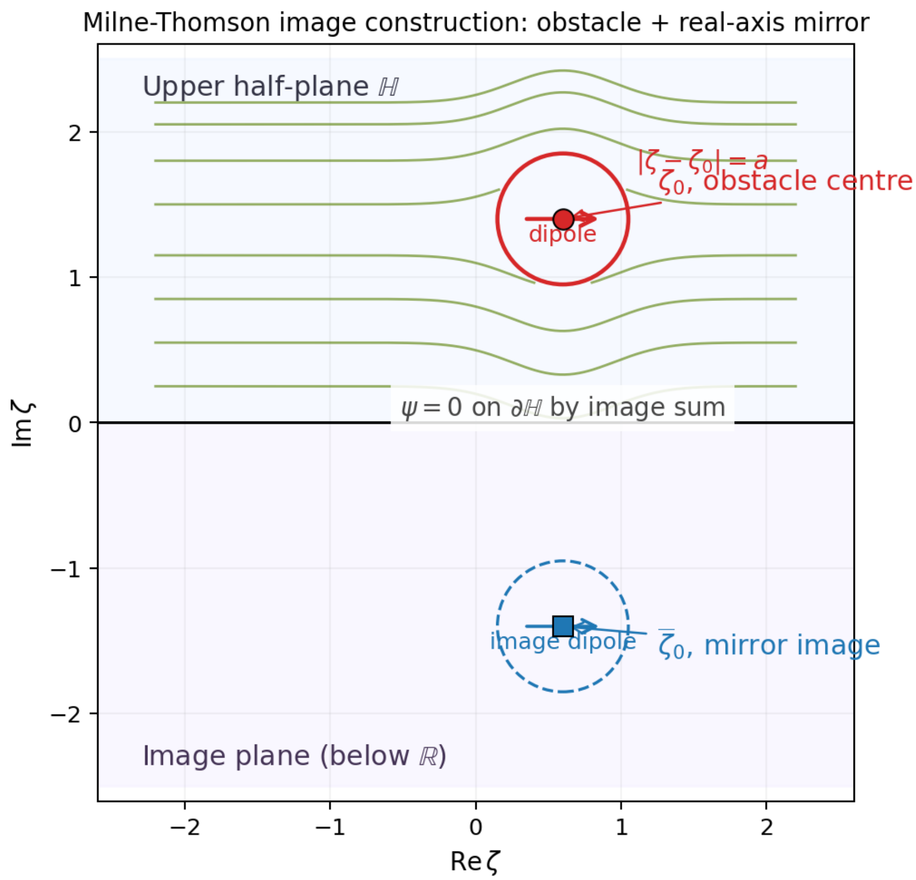


Figure 1: **Milne-Thomson image construction in $\mathbb{H}$.** Red: obstacle centre ζ_0 and its circle $|\zeta - \zeta_0| = a$. Blue: real-axis mirror $\overline{\zeta_0}$ and mirror circle (dashed) in the lower half-plane. The dipole direction is indicated by horizontal arrows at each centre. Schematic streamlines (green) show how the dipole sum simultaneously enforces $\psi = \mathrm{const}$ on $|\zeta - \zeta_0| = a$ and $\psi = 0$ on $\partial\mathbb{H}$.

2.8 SC Crowding and the Conformal Modulus

A practical obstruction to numerical SC is *crowding*: for polygons with strongly different edge-length scales, pre-vertices cluster exponentially on the real axis. The phenomenon reflects a rigid complex-analytic invariant. For a rectangle of width $2K(k)$ and height $K'(k)$, with K, K' the complete elliptic integrals [Olver et al., 2010, §19.2], the conformal modulus $m(R) = 2K(k)/K'(k)$ is a conformal invariant. The asymptotic expansion of $K(k)$ [Olver et al., 2010, §19.12] gives inner pre-vertex spacing

$$1/k - 1 \;\sim\; 8e^{-\pi m(R)/2} \qquad \text{as } m(R) \to \infty. \tag{12}$$

For a general polygon containing a long thin region of aspect ratio L, spacing across that region contracts exponentially in L [Driscoll and Trefethen, 2002, Ch. 2]. Figure 2 shows pre-vertex spacings recovered by our solver tracking the theoretical curve until the $O(N^{-1})$ Gauss–Legendre branch-point error dominates ($\sim 10^{-3}$ to 10^{-4}). Spacing falls below 10^{-8} at $L \approx 7$ and beyond $L \gtrsim 10$ approaches double precision, limiting practical SC solvers to $n \lesssim 20$ vertices with mild aspect ratios. The 2752-vertex Boulder boundary must therefore be simplified to 12 vertices (Section 3.1) with all interior angles in $[0.35\pi, 1.75\pi]$.

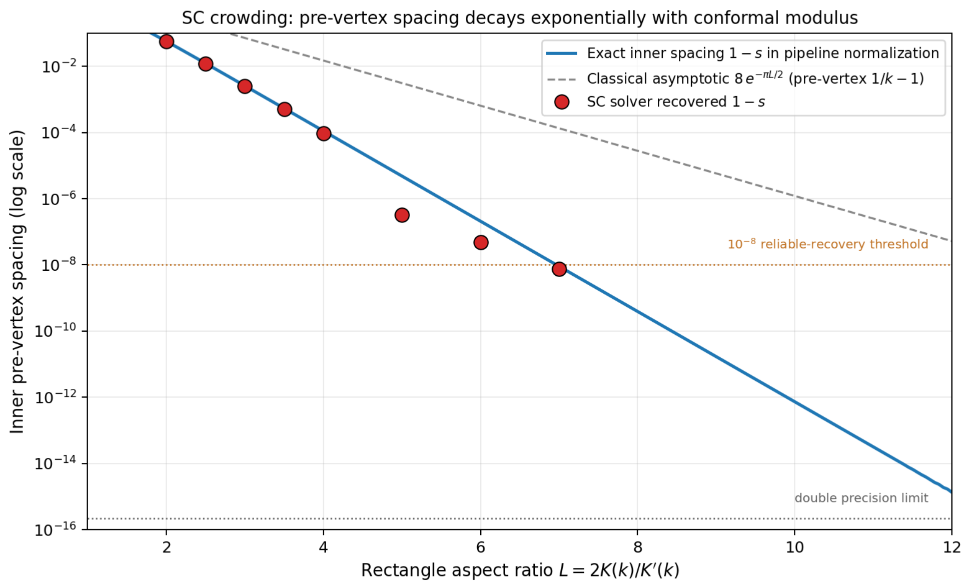


Figure 2: **SC crowding: pre-vertex spacing as a function of rectangle aspect ratio $L = 2K(k)/K'(k)$.** Solid blue: exact inner spacing $1 - s$ in the pipeline normalisation $\{-1, 0, s, 1\}$, where $s = 2k/(1 + k^2)$ by (18). Dashed grey: classical asymptotic $8e^{-\pi L/2}$ for $1/k - 1$ in the symmetric normalisation. Red circles: spacings recovered by our SC solver on rectangles of aspect ratio $L \in \{1.5, 2, 2.5, 3, 3.5, 4, 5, 6, 7\}$; they track the theoretical curve until the quadrature error of plain Gauss–Legendre at the branch-point endpoints ($\sim 10^{-3}$–10^{-4}) dominates. The dashed orange line at 10^{-8} marks the threshold below which the pre-vertex spacing falls within the LM solver's default tolerance and can no longer be reliably recovered.

3 Numerical Implementation

3.1 Polygon Preprocessing

The raw Boulder County TIGER/Line boundary [US Census Bureau, 2025] has 2752 vertices. Douglas–Peucker simplification with an adaptive binary-search tolerance reduces it to 14 vertices; two vertices with excessively acute angles below 0.35π are then removed iteratively by merging adjacent edges, yielding the final $n = 12$ polygon. The polygon is normalised to $O(1)$ scale and centred at the origin for numerical stability. The angle bounds $[0.35\pi, 1.75\pi]$ are chosen from the crowding discussion of Section 2.8.

3.2 The SC Parameter Problem

The $n - 3 = 9$ free pre-vertices $\zeta_2, \ldots, \zeta_{n-2} \in (0, 1)$ are chosen so that the image polygon $f(\mathbb{R})$ has the correct side-length ratios. Define the k-th side length

$$L_k = \left| \int_{\zeta_k}^{\zeta_{k+1}} \prod_{j=1}^{n} (t - \zeta_j)^{\alpha_j - 1} \, \mathrm{d}t \right|, \tag{13}$$

and require

$$\frac{L_k}{L_{n-1}} = \frac{|w_{k+1} - w_k|}{|w_0 - w_{n-1}|}, \qquad k = 0, \ldots, n-2. \tag{14}$$

Softmax reparameterisation. Given $p \in \mathbb{R}^{n-3}$, define weights and cumulative pre-vertex positions:

$$w_j(p) = \frac{e^{p_j}}{\sum_{i=1}^{n-3} e^{p_i} + 1}, \qquad \zeta_{k+1} = \sum_{j=1}^{k} w_j(p). \tag{15}$$

Since $e^{p_j} > 0$, each $w_j > 0$ and the cumulative sums are strictly increasing. Let $S = \sum_i e^{p_i}$; then $\sum_j w_j = S/(S+1) < 1$, so every $\zeta_k \in (0, 1)$. The ordering and box constraints are thus automatic, and the optimisation over $p \in \mathbb{R}^{n-3}$ is unconstrained.

Solver and quadrature. System (14) is solved by Levenberg–Marquardt least squares [Stoer and Bulirsch, 2002, Ch. 7] via SciPy's `least_squares` routine. Each side-length integral is evaluated by $N = 500$-node Gauss–Legendre quadrature [Olver et al., 2010, §3.5(v)][Dahlquist and Björck, 2008] on $[-1, 1]$ after affine rescaling to $[\zeta_k, \zeta_{k+1}]$; nodes and weights are pre-computed and reused across iterations, so each evaluation is a single vectorised dot product. Levenberg–Marquardt update details are in Appendix A.

Figure 3 shows the recovered pre-vertex layout for the 12-vertex Boulder County polygon alongside the polygon itself, with each $\zeta_k \leftrightarrow w_k$ pair colour-matched. The concentration of pre-vertices toward one side of the real axis directly visualises the crowding phenomenon of Section 2.8: vertices on the narrow western flank of the polygon share their pre-vertex region with several close neighbours.

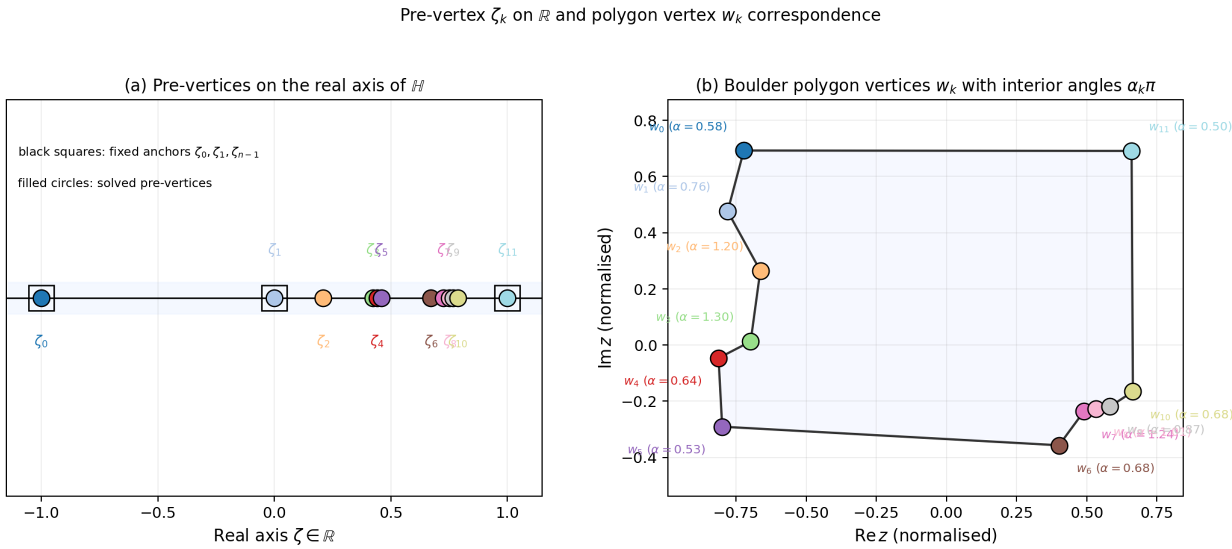


Figure 3: **Pre-vertex $\zeta_k \leftrightarrow$ polygon vertex w_k correspondence.** Matching colours identify each pair. Black squares in panel (a) mark the three Möbius-fixed anchors $\zeta_0 = -1, \zeta_1 = 0, \zeta_{n-1} = 1$; the nine filled circles are free pre-vertices recovered by the solver. Panel (b) gives each vertex with its interior angle $\alpha_k \pi$. Clustering of pre-vertices in $[0, 0.5]$ reflects the asymmetric edge-length distribution of the simplified Boulder boundary.

Determining A **and** C**.** Once the pre-vertices are known, $A, C \in \mathbb{C}$ are obtained from the overdetermined linear system $A + C \cdot I_k = w_k$ for $k = 0, 1, \ldots, n-1$, where I_k is the SC integral from a reference point $\zeta_{\text{ref}} = 0.5i$ to $\zeta_k + \varepsilon$ (with a small complex offset ε to avoid the branch point). The least-squares solution is computed via `numpy.linalg.lstsq`.

3.3 Forward Map and Streamline Computation

All flow results are produced via the *forward-map* approach. A pre-vertex-adapted grid of ζ values in $\mathbb{H}$ (dense samples between every adjacent pre-vertex pair, 60 logarithmically spaced imaginary parts) is mapped forward to $z = f(\zeta)$, each SC integral evaluated by a 250-node Gauss–Legendre rule along an L-shaped path in $\mathbb{H}$ that clears the real-axis branch cuts (Figure 4). The complex potential $W(\zeta)$ is evaluated analytically at each ζ point. The resulting scattered pairs $(z_i, W(\zeta_i))$ are interpolated onto a regular physical-plane grid (`scipy.interpolate.griddata`, linear method), and contours of $\operatorname{Im} W$ (streamlines) and $\operatorname{Re} W$ (equipotentials) are extracted.

This avoids the per-pixel Newton iteration that inverse mapping would require. Grid points from $\mathbb{H}$ land strictly inside Ω by construction, so the interpolated field is zero outside Ω with smooth contours at the boundary.

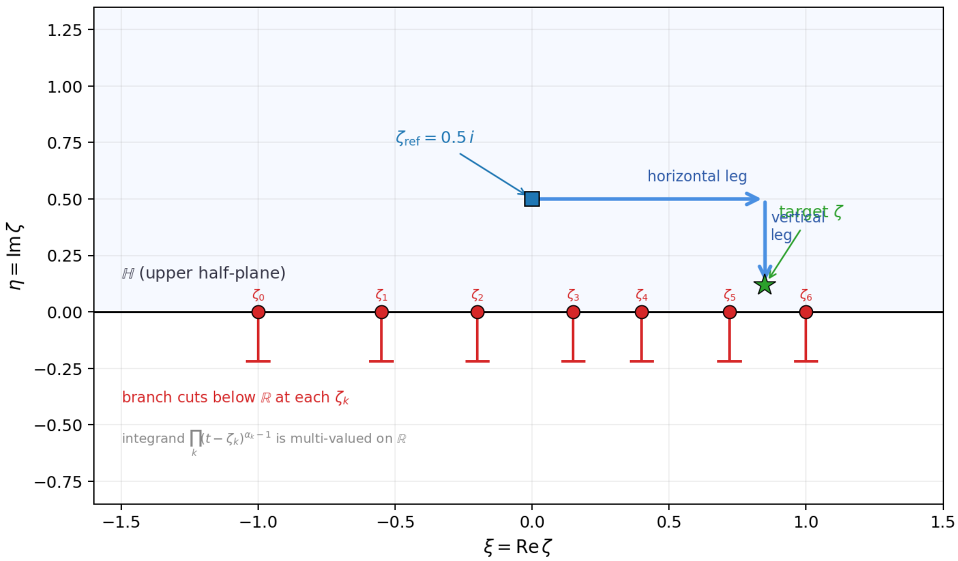


Figure 4: **L-shaped integration path used in the forward SC map.** Red markers on the real axis are the pre-vertices ζ_k; short red segments below $\mathbb{R}$ indicate the branch cuts of the integrand $\prod_k (t-\zeta_k)^{\alpha_k-1}$. The blue square is the reference point $\zeta_{\mathrm{ref}} = 0.5i$; the green star is a target ζ in $\mathbb{H}$. The path is taken horizontally within $\mathbb{H}$ to the target's real part (horizontal leg at $\mathrm{Im}\,\zeta = 0.5$, passing above the pre-vertices on $\mathbb{R}$), then vertically to the target. Both legs stay strictly above the real axis, so they never cross a branch cut (which extends downward from each ζ_k into the lower half-plane), preserving the chosen branch of the SC integrand.

3.4 Verification on the Rectangle Map

Before applying the pipeline to Boulder, we verify it against the one polygonal case with a closed-form SC map: the rectangle. The classical construction [Nehari, 1952, Ch. V, §6][Ahlfors, 1979, Ch. 6, §2.3] uses the symmetric pre-vertex set $\{-1/k, -1, 1, 1/k\}$ with elliptic modulus $0 < k < 1$, giving

$$f(\zeta) = \int_0^\zeta \frac{\mathrm{d}t}{\sqrt{(1-t^2)(1-k^2t^2)}} = F(\arcsin\zeta,\, k), \tag{16}$$

with F the incomplete elliptic integral of the first kind [Olver et al., 2010, §19.2]. The corner images are $f(\pm 1) = \pm K(k)$ and $f(\pm 1/k) = \pm K(k) + iK'(k)$, so the image rectangle has width $2K(k)$ and height $K'(k)$; its conformal modulus matches Section 2.8.

Adaptation to the project's parameterisation. The softmax reparameterisation (15) constrains all free pre-vertices to $(0, 1)$ and fixes three anchors at $\{-1, 0, 1\}$. Values $\pm 1/k$ with $k < 1$ lie outside this range, so the classical symmetric placement cannot be used directly. The two normalisations are related by the unique real Möbius transformation matching the ordered cross-ratio on the real axis:

$$[-1, 0; s, 1] = [-1/k, -1; 1, 1/k], \tag{17}$$

where $[a, b; c, d] = (a-c)(b-d)/[(a-d)(b-c)]$. Computing both sides,

$$\frac{1+s}{2s} = \frac{(1+k)^2}{4k} \quad\Longrightarrow\quad s(1+k)^2 = 2k(1+s) \quad\Longrightarrow\quad s = \frac{2k}{1+k^2}. \tag{18}$$

This gives a one-to-one correspondence between $k \in (0, 1)$ in the classical picture and the single free pre-vertex $s \in (0, 1)$ in our pipeline. For a target rectangle of conformal modulus $m(R) = 2K(k)/K'(k)$, the pipeline must recover the corresponding s determined by (18).

Numerical verification. Take $m(R) = 2$ (the standard 2×1 rectangle), giving $k = 1/\sqrt{2}$ and $s_{\text{exact}} = 2\sqrt{2}/3 \approx 0.94281$. The softmax + Levenberg–Marquardt solver converges to $s_{\text{recovered}} = 0.94296$, with absolute error $\approx 1.5 \times 10^{-4}$ on the pre-vertex and side-length residual $\|r\|_2 \approx 7 \times 10^{-5}$. The accuracy floor reflects $O(N^{-1})$ Gauss–Legendre convergence at the integrable $(t - \zeta_k)^{-1/2}$ branch-point endpoints [Dahlquist and Björck, 2008]; a Duffy substitution $t - \zeta_k = u^2$ would absorb the singularity and recover near machine precision at the cost of re-working the quadrature module. The current $\sim 10^{-4}$ accuracy is sufficient for the qualitative Boulder results of Section 4. Figure 5 shows the configuration and LM convergence history.

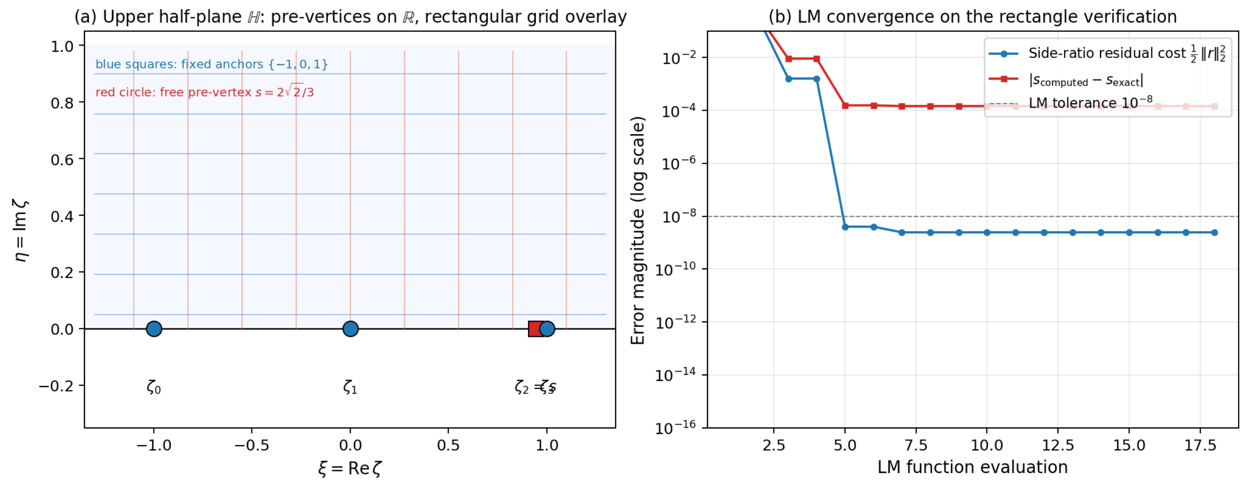


Figure 5: **Rectangle verification of the SC pipeline.** Target: $m(R) = 2$ rectangle, exact $s = 2\sqrt{2}/3 \approx 0.94281$; recovered $s \approx 0.94296$. (a) Upper half-plane $\mathbb{H}$ with the pipeline's pre-vertex layout: blue squares are the three Möbius-fixed anchors $\{-1, 0, 1\}$; the red circle is the free pre-vertex s at its exact position for $k = 1/\sqrt{2}$. A rectangular reference grid on $\mathbb{H}$ is overlaid to illustrate the conformal-grid image structure used in Section 4. (b) Levenberg–Marquardt convergence: the side-length residual cost $\frac{1}{2}\|r\|_2^2$ plateaus near 3×10^{-9} while the pre-vertex error $|s_{\text{computed}} - s_{\text{exact}}|$ plateaus near 1.5×10^{-4}; the two curves stall at different floors, consistent with the $O(N^{-1})$ quadrature convergence rate at branch-point endpoints.

4 Application to the Boulder Polygon

4.1 Polygon Simplification

Figure 6 shows the original 2752-vertex TIGER/Line boundary of Boulder County alongside the 12-vertex polygon used in the SC computation. Douglas–Peucker removes fine-scale boundary undulations; the interior-angle filter then enforces $\alpha_k \in [0.35\pi, 1.75\pi]$. The aspect ratio and dominant geometric features are faithfully preserved.

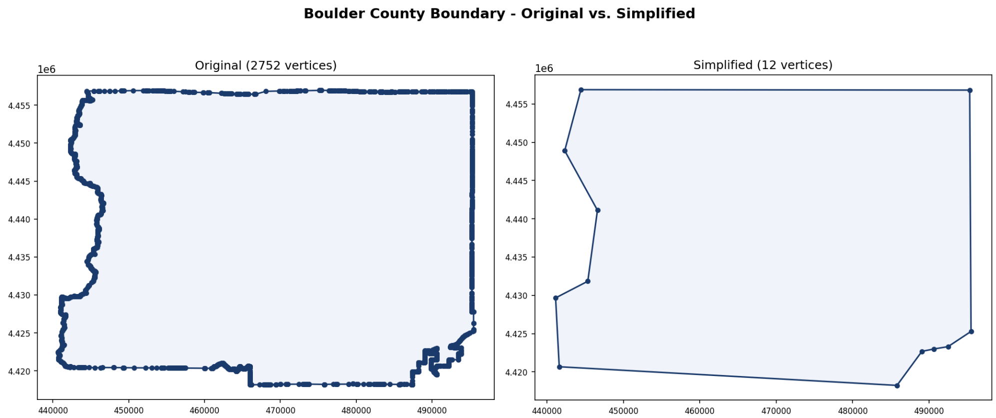
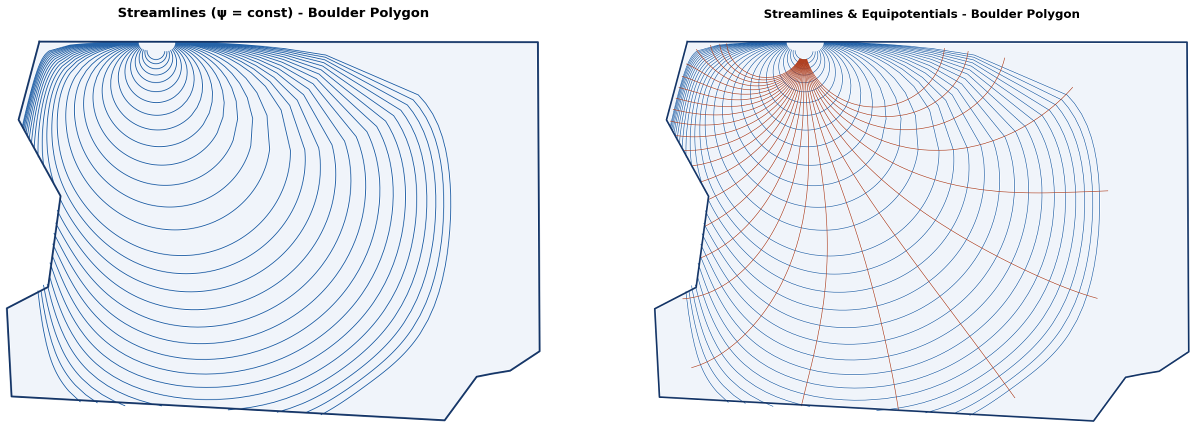


Figure 6: **Boulder boundary before and after SC preprocessing.** Left: original 2752-vertex TIGER/Line polygon [US Census Bureau, 2025]. Right: 12-vertex simplification after Douglas–Peucker reduction and iterative angle filtering with bounds $\alpha_k \in [0.35\pi, 1.75\pi]$. The simplified polygon is the domain Ω for all subsequent SC computations.

4.2 Uniform Ideal Flow

Topological note. Uniform flow streamlines $\operatorname{Im} \zeta = \eta_0$ are horizontal lines in $\mathbb{H}$ running from $-\infty$ to $+\infty$, which on the Riemann sphere meet at a single point at infinity. The SC map sends that point to one fixed vertex on $\partial\Omega$, so every streamline becomes a closed curve in Ω. The circulatory appearance of Figure 7 is therefore intrinsic to mapping $\mathbb{H}$ onto a bounded polygon. To obtain crossing streamlines, replace $W = U\zeta$ with a source-sink pair $W = U \log[(\zeta - \zeta_{\text{src}})/(\zeta - \zeta_{\text{sink}})]$.

Uniform flow results. Figure 7 presents the uniform flow $W = U\zeta$ in Ω. Streamlines $\psi = \text{const}$ in panel (a) stay inside $\partial\Omega$, satisfying the no-penetration condition exactly by construction. Panel (b) overlays streamlines and equipotentials; their orthogonality throughout the interior is the definitive visual confirmation of conformality, a consequence of the Cauchy–Riemann equations (1) wherever $f'(\zeta) \neq 0$.



(a) Streamlines $\psi = \text{const}$ (blue).

(b) Conformal grid (streamlines + equipotentials).

Figure 7: **Uniform flow $W = U\zeta$ in the 12-vertex Boulder County polygon Ω.** Streamlines are the forward images of horizontal lines $\text{Im}\zeta = \eta_0$ in $\mathbb{H}$; equipotentials are images of $\text{Re}\zeta = \xi_0$. The SC map $f : \mathbb{H} \to \Omega$ preserves angles locally, so the two families meet orthogonally throughout the interior. Pre-vertex-adapted ζ-grid mapped to an 80×80 physical grid; 28 forward-mapped curves per family. Closed-streamline appearance is a topological consequence discussed above.

4.3 Doubly-Connected Flow with an Urban Obstacle

The downtown commercial core (hardcoded downtown Boulder polygon, $9.2\,\text{km}^2$, clipped inside the county boundary) is mapped to $\mathbb{H}$ via f^{-1}; its pre-image is enclosed by a minimum enclosing circle (Welzl's algorithm [Welzl, 1991]) of radius a centred at $\zeta_0 \in \mathbb{H}$. The urban-obstacle potential (11) then applies, with achieved separation ratio $\text{Im}(\zeta_0)/a \approx 3.96$ for the Boulder County run, giving a leading-order term of size $O\big((a/\text{Im}\,\zeta_0)^2\big) \approx 6.4\%$.

Figure 8 shows the result. The Milne-Thomson dipole at ζ_0 enforces no-penetration on the enclosing circle; the image term at $\overline{\zeta_0}$ restores $\psi = 0$ on $\mathbb{R}$. Streamlines deflect visibly around the purple obstacle, with local acceleration on its flanks.

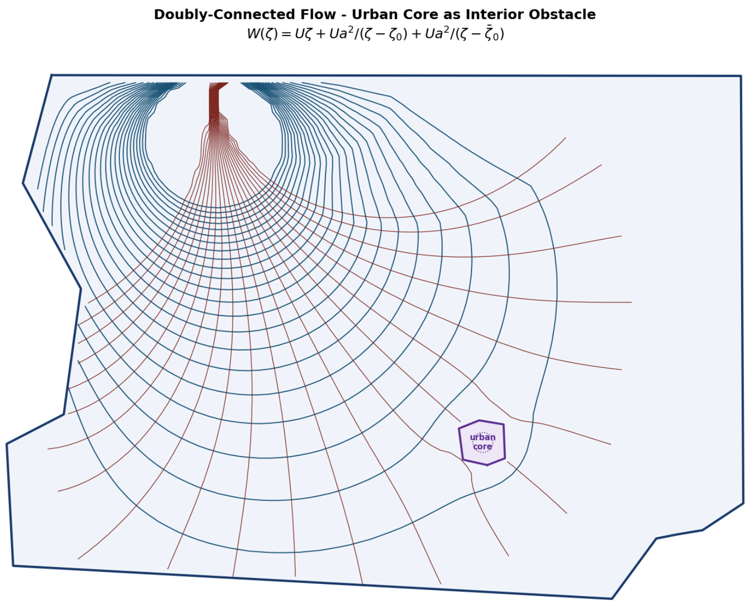


Figure 8: **Doubly-connected flow with urban-core obstacle via the Milne-Thomson circle theorem** (11)**.** Downtown Boulder commercial core shown in purple. Separation ratio $\mathrm{Im}(\zeta_0)/a \approx 3.96$; leading-order term of size $\approx 6.4\%$. Streamlines deflect around the obstacle, with a local acceleration on its flanks. Same grid resolution and contour counts as Figure 7.

Velocity magnitude. Figure 9 shows $|W'(z)|$ for both flow models as heatmaps. In the uniform case, the speed peaks at the polygon vertices (the SC map has $f'(\zeta_k) = 0$ singularities, so $|W'(z)| = U/|f'(\zeta)| \to \infty$ as $z \to w_k$), giving the expected corner-acceleration signature. In the urban-obstacle case, the dipole term additionally compresses the flow on the flanks of the downtown core, producing two localised acceleration regions north and south of the obstacle that the streamline figure only hints at qualitatively.

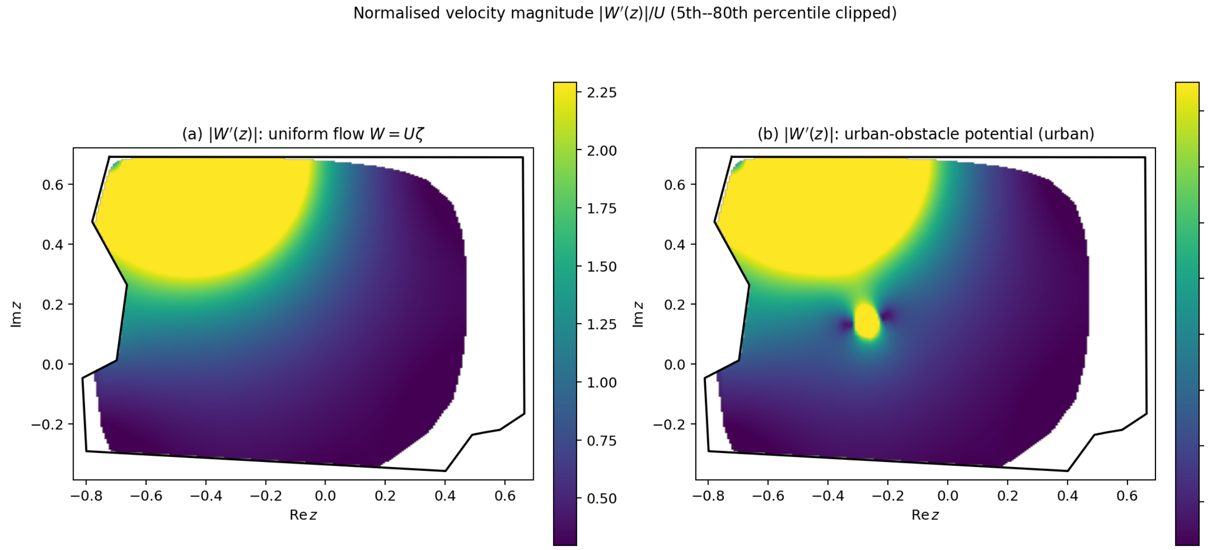


Figure 9: **Normalised velocity magnitude $|W'(z)|/U$ for the two flow models.** (a) Uniform flow: speed rises sharply near polygon vertices where $f'(\zeta_k) \to 0$. (b) Urban-obstacle potential: two additional bright regions on the flanks of the downtown core quantify the "local acceleration" visible in Figure 8. Colour bar clipped to the 5th–80th percentile to suppress the finite-grid blow-up at vertex branch points while keeping the flank acceleration visible.

Remark on modularity. Comparing Figures 7(a) and 8, both panels share the same SC map $f : \mathbb{H} \to \Omega$ and the same normalised polygon, so every visible difference comes from the analytic form of $W(\zeta)$ alone (uniform flow vs. the urban-obstacle dipole (11)). The SC map is computed once; the physical model is switched by evaluating a different closed-form expression in $\mathbb{H}$.

5 Conclusions

The Schwarz–Christoffel transformation provides an exact, closed-form analytic framework for ideal fluid flow in a simply connected polygonal domain. The forward-map approach evaluates analytically tractable potentials in $\mathbb{H}$ and maps the results to Ω, yielding streamlines that satisfy the no-penetration condition exactly by construction. Two complex potentials were derived in detail and visualised on the Boulder polygon: uniform flow and a doubly-connected model with a Milne-Thomson interior obstacle. A closed-form verification on the rectangle-to-half-plane map (an elliptic-integral identity) confirmed the numerical pipeline to within the expected quadrature-limited accuracy ($\sim 10^{-4}$).

The principal mathematical content is the interplay between conformal-mapping theory (Riemann Mapping Theorem, SC integral formula, interior-angle structure, Möbius normalisation, Schwarz reflection) and ideal-flow theory (complex potential, method of images, Milne-Thomson circle theorem). The SC formula (4) encodes polygon geometry through the exponents $\alpha_k - 1$: each vertex angle becomes a branch-point strength. The Cauchy–Riemann equations, through the conformality of f, guarantee that every solution of Laplace's equation in $\mathbb{H}$ pulls back to a solution in Ω.

A practical obstruction identified in Section 2.8 is *SC crowding*: the pre-vertex spacing across a long region of aspect ratio L contracts exponentially in L, a rigid conformal invariant tied to the modulus of the region. This forces the 2752-vertex Boulder County boundary to be simplified to 12 vertices before any SC computation is feasible.

Limitations and scope. Three limitations bound the present pipeline. First, the Milne-Thomson approximation (11) carries a leading-order error of $O((a/\operatorname{Im}\zeta_0)^2) \approx 6.4\%$ for the Boulder run (Section 4.3). For qualitative streamline visualisation this is acceptable; for quantitative pressure or drag calculations it is not. Any application requiring rigorous force coefficients must replace the dipole approximation with an exact doubly-connected SC map, as noted in the first future direction below. Second, the exponential crowding described in Section 2.8 caps the practical polygon complexity at $n \lesssim 20$ vertices. A Manhattan-scale urban boundary, with hundreds of geometrically distinct blocks, lies far outside this range. Treating such domains would require either domain decomposition into patches each amenable to a separate SC map, or a cross-ratio reformulation of the parameter problem along the lines of Driscoll and Trefethen [Driscoll and Trefethen, 2002, Ch. 4] that reduces crowding sensitivity. Third, the forward-map interpolation used in Section 3.3 degrades near polygon vertices where $f'(\zeta_k) \to 0$. This is an intrinsic property of the SC branch-point structure, not a numerical artefact: the physical velocity $|W'(z)| = U/|f'(\zeta)|$ diverges at each corner, so a uniform interpolation grid cannot resolve the local field accurately. Any application requiring precise flow near corner singularities would need asymptotic local expansions rather than the griddata interpolation used here.

Future directions. Two extensions are natural.

- **Exact doubly-connected SC map.** The Milne-Thomson approximation (11) introduces a small but non-zero approximation error (see Section 4.3). An exact doubly-connected SC map [Driscoll and Trefethen, 2002, Ch. 4, §4.9], implemented via the Schottky double or a circular slit canonical region, would eliminate this approximation and give a rigorous treatment.
- **Multiply connected domains.** The formalism extends to domains with several obstacles via the Schottky-double conformal map, enabling an SC-style treatment of full building footprints rather than a single enclosing circle.

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A Levenberg–Marquardt Solver Details

Given the residual vector $r(p) \in \mathbb{R}^{n-1}$ of (14) and Jacobian $J = \partial r / \partial p$, Levenberg–Marquardt updates

$$p^{(t+1)} = p^{(t)} - \left(J^\top J + \lambda_t \operatorname{diag}(J^\top J)\right)^{-1} J^\top r(p^{(t)}), \tag{19}$$

where λ_t is an adaptive damping parameter. When $\lambda_t \to 0$ the update reduces to the Gauss–Newton step (fast second-order convergence near a good solution); when $\lambda_t \to \infty$ it reduces to a small gradient-descent step (stable near ill-conditioned regions). The damping is increased whenever a proposed step fails to reduce $\|r\|_2$ and decreased on successful steps. For the Boulder County 12-vertex problem the solver reduces the LM cost to $\approx 5.4 \times 10^{-2}$ after seven restart passes. An optimal cyclic rotation of the vertex ordering (roll $= 5$) is applied before the solve to place the longest polygon side as the outer-arc normalisation term, keeping all target side-length ratios below unity and reducing the maximum ratio from ≈ 31 to ≈ 0.87. The remaining cost reflects mild crowding of four free pre-vertices into two sub-intervals of width ≈ 0.02 near $\zeta \approx 0.44$ and $\zeta \approx 0.76$.

B Code and Reproducibility

All source code is available at the project GitHub repository[1]. The two-model pipeline is reproduced by:

```
pip install -r requirements.txt
python main.py --shapefile data/raw/tl_2025_08_county --urban --grid 80
```

Key modules: `src/polygon.py` (boundary preprocessing), `src/angles.py` (interior-angle computation), `src/sc_solver.py` (SC parameter problem, Gauss–Legendre quadrature, raw SC integral), `src/flow.py` (forward-map streamlines, scattered interpolation onto physical-plane grid), `src/urban.py` (OSM obstacle, Welzl minimum enclosing circle), `src/sc_solver_dc.py` (doubly-connected potentials), and `src/visualization.py` (figure generation).

Software: Python 3.14, NumPy 2.4, SciPy 1.17, Shapely 2.1, GeoPandas 1.1, OSMnx 2.1 [Boeing, 2017], Matplotlib 3.10.

[1] https://github.com/McDonaldCrispyThigh/Ideal-Flow-SC